

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 919+580

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

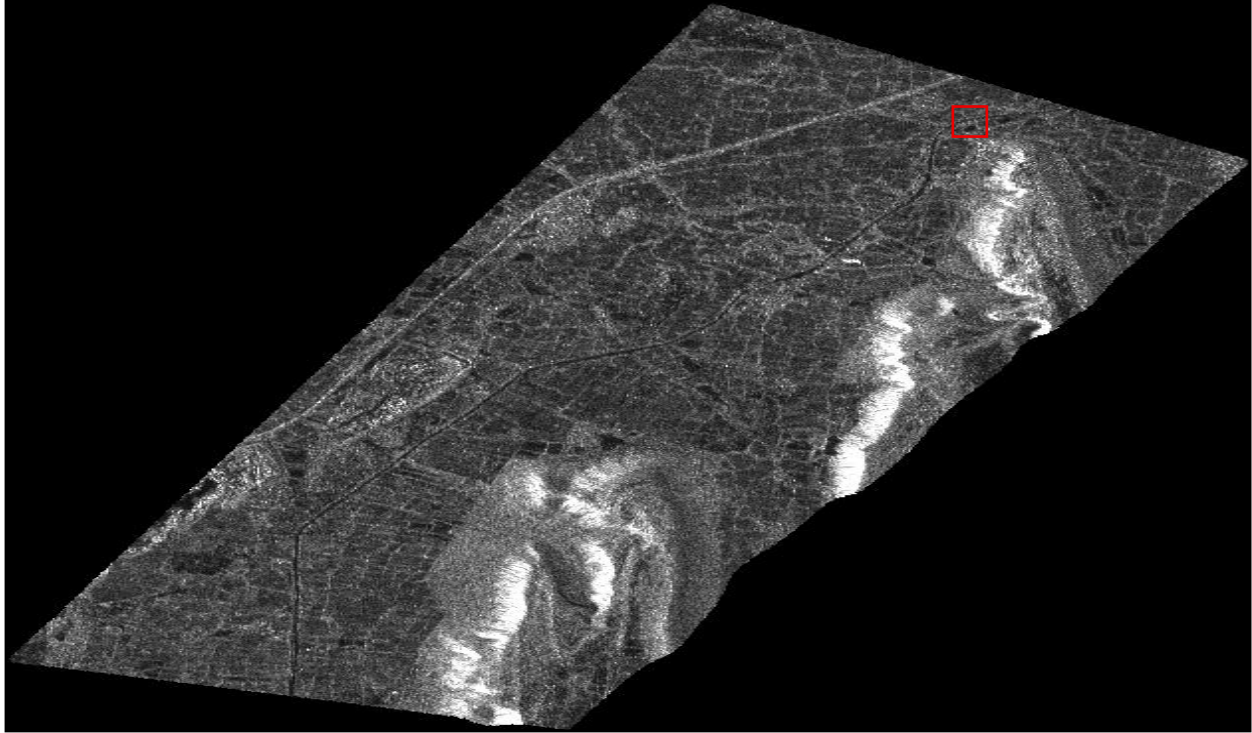
Flight Direction - DESCENDING

Polarization – VV+VH

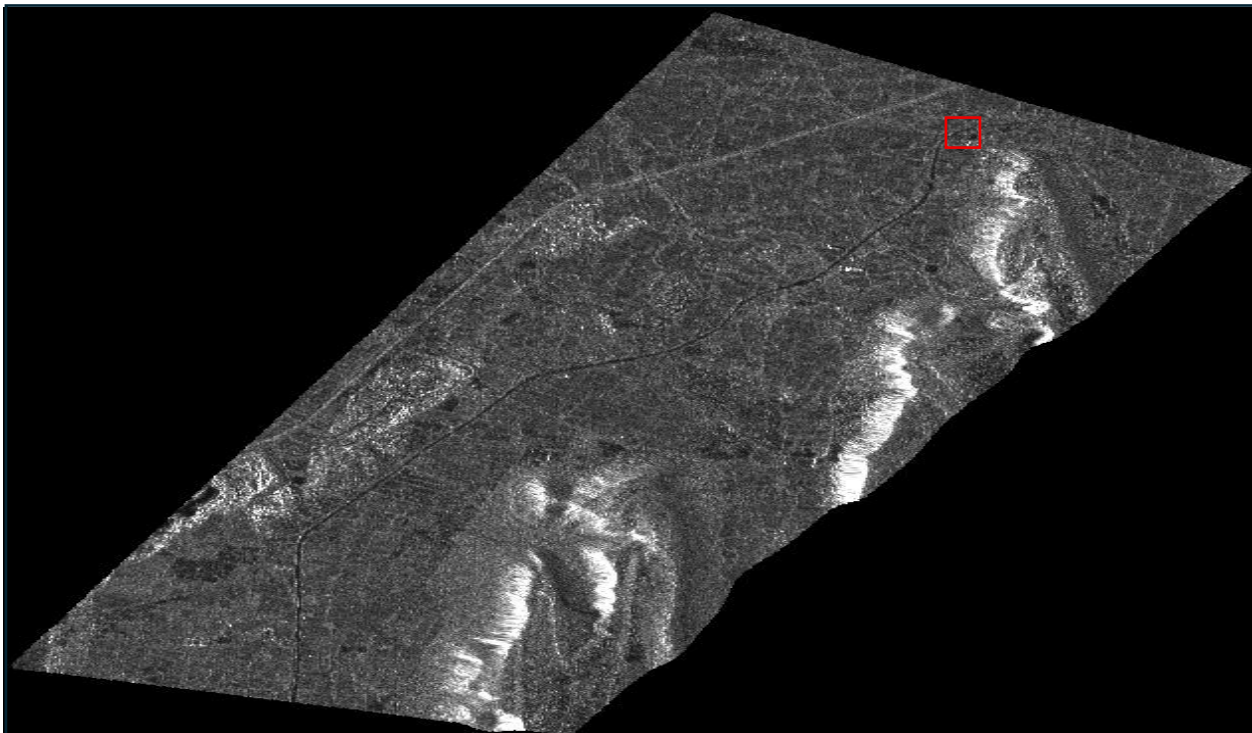
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

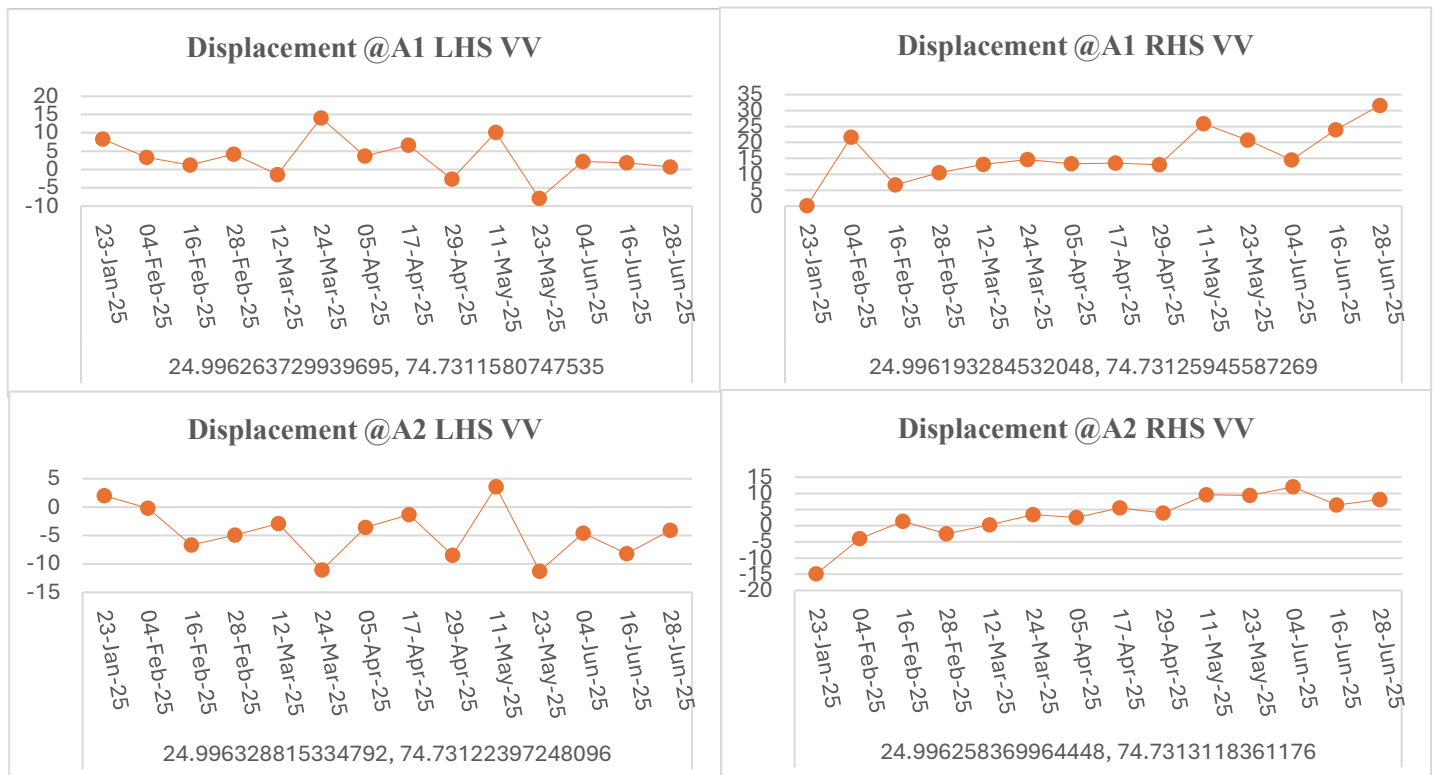
VUP – CH-917+050

Length – 21.9M

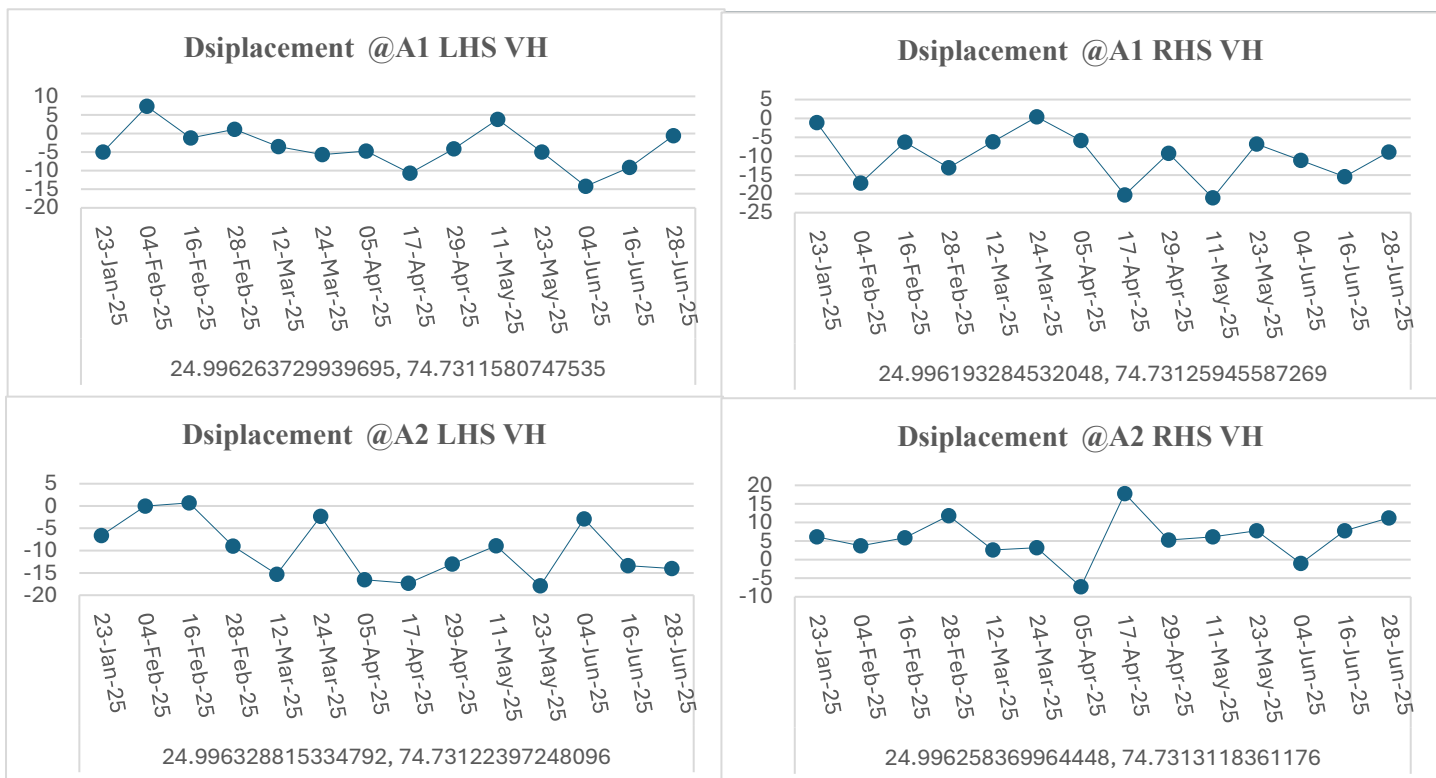


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



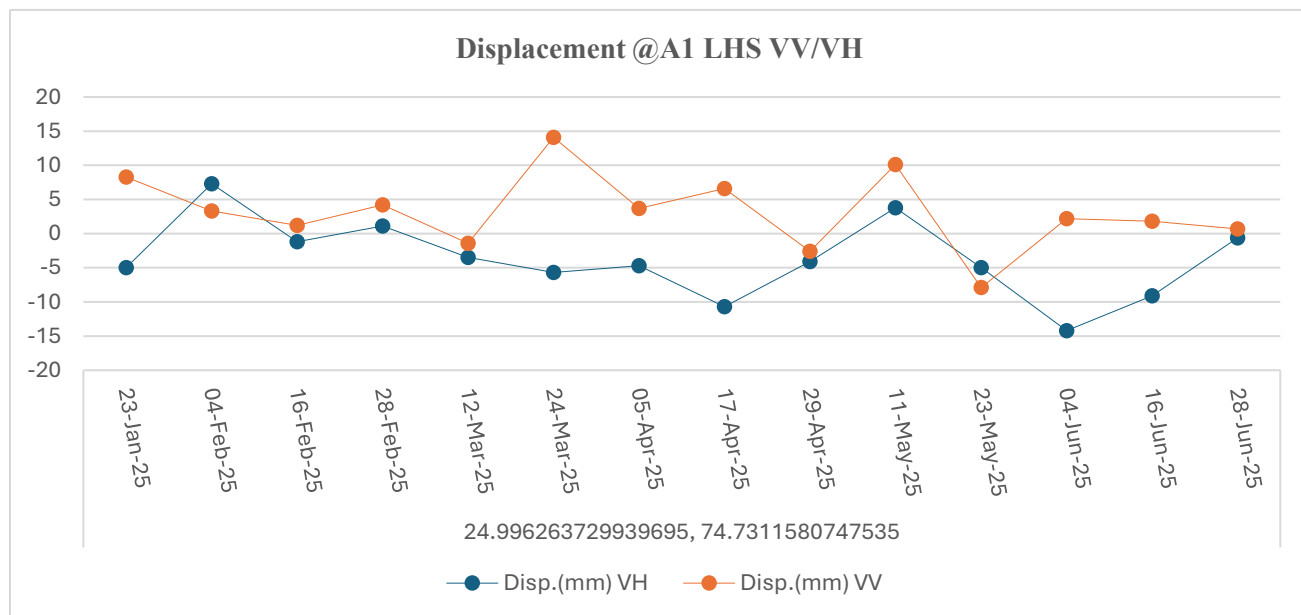
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.996263729939695, 74.7311580747535)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** VV shows wider fluctuations, approximately between -8 mm and +14 mm.
- **Horizontal Displacement (VH)** VH exhibits moderate variability, generally ranging from about -14 mm to +8 mm.

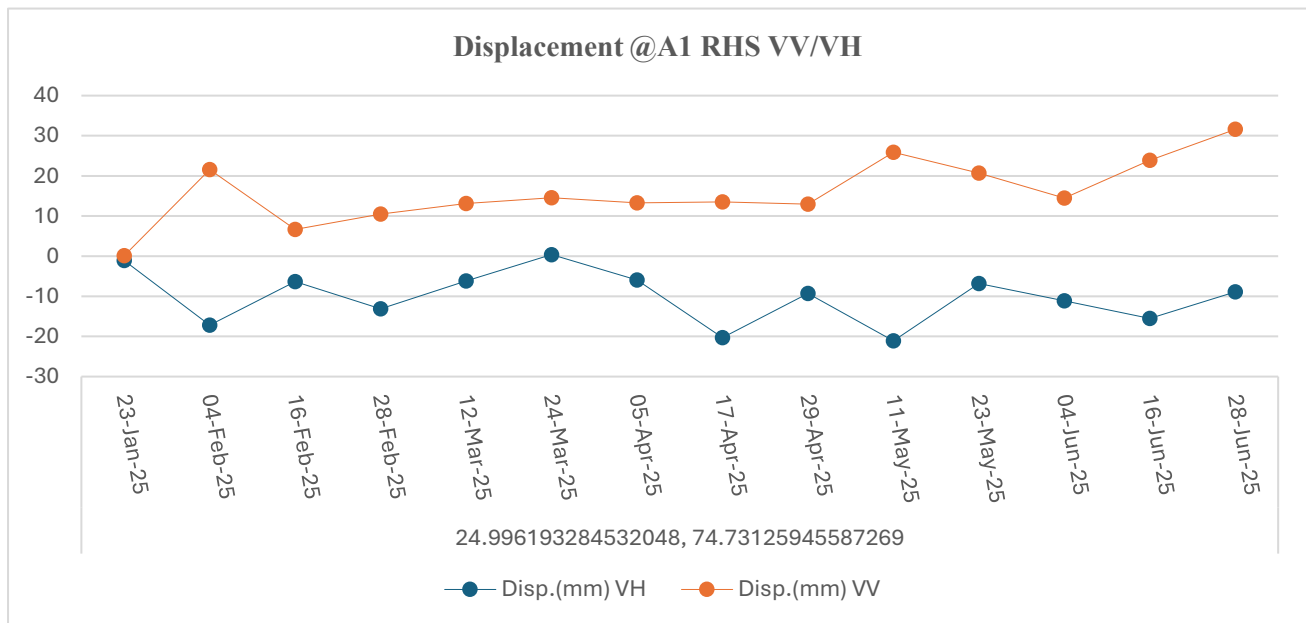
Critical Events & Anomalies:

- **February 2025:** Both VH and VV approach near-zero to slightly positive values, indicating a briefly balanced structural response.
- **May 2025:** VV again shows a secondary uplift peak (+10 mm) around 11-May-25, accompanied by a small positive VH (+4 mm), indicating coincident but limited lateral response.
- **June 2025:** while VV turns slightly negative (~-6 mm), marking a temporary settlement-dominated phase. Both VH and VV recover toward near-zero to small positive values, suggesting partial stabilization and recovery.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.996193284532048, 74.73125945587269)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** VH remains predominantly negative throughout the monitoring period, generally ranging from approximately –22 mm to 0 mm.
- **Vertical displacement (VV)** VV exhibits consistently positive displacement, varying roughly between +5 mm and +32 mm.

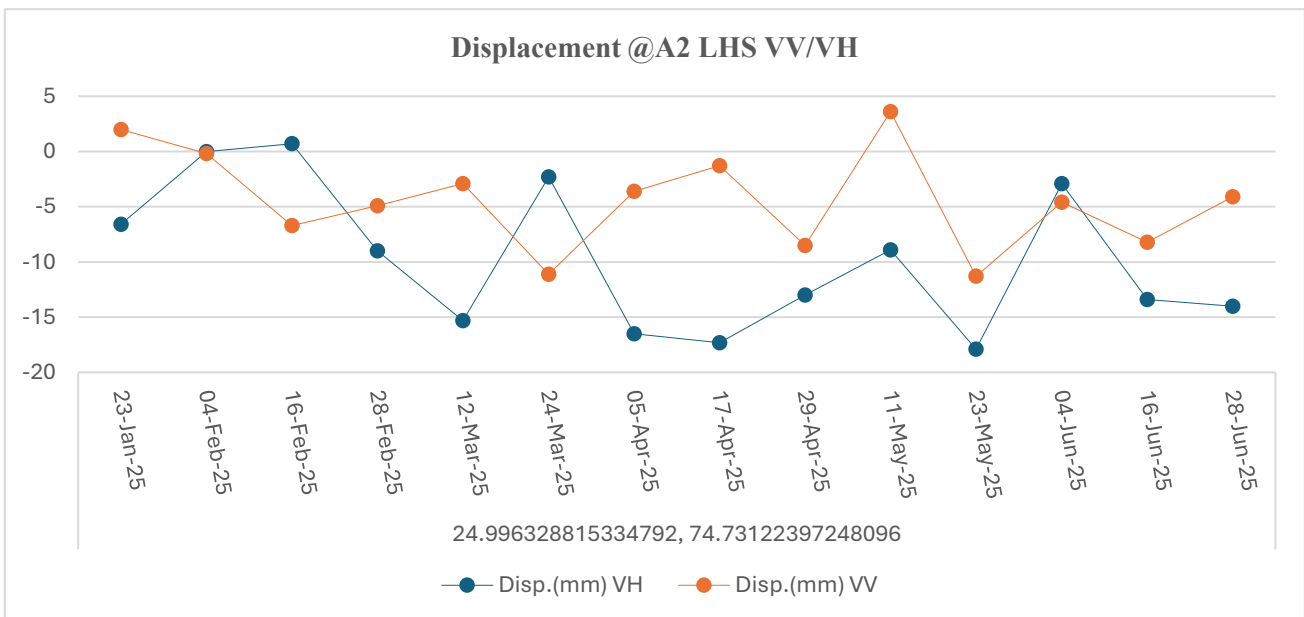
Critical Events & Anomalies:

- **February 2025:** VV shows a sharp uplift peak while VH drops to around –15 mm, indicating strong vertical response decoupled from horizontal movement.
- **March 2025:** VH briefly approaches near-zero displacement, coinciding with VV around +14 mm, indicating a temporary reduction in lateral bias.
- **June 2025:** VV attains the highest recorded uplift by 28-Jun-25, while VH remains moderately negative (–10 mm), indicating sustained vertical accumulation with persistent lateral bias.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.996328815334792, 74.73122397248096)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** VH remains consistently negative across the entire monitoring period, typically ranging from approximately –2 mm to –18 mm.
- **Vertical Displacement (VV)** VV exhibits small-magnitude fluctuations around the zero line, varying roughly between –12 mm and +4 mm.

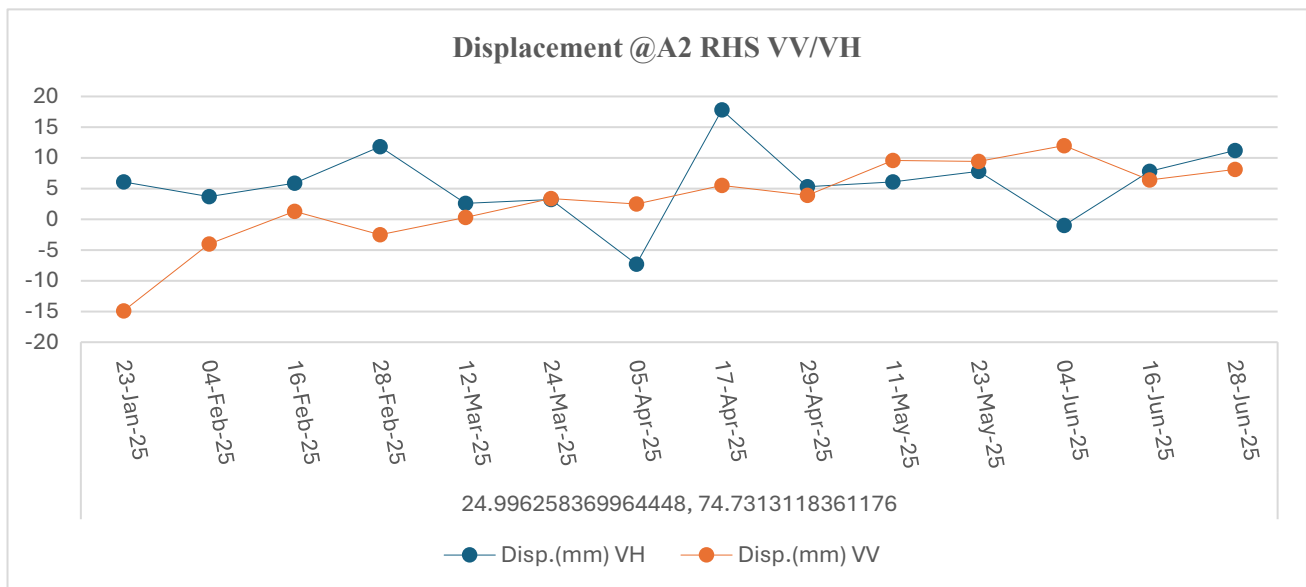
Critical Events & Anomalies:

- **February 2025:** VV briefly approaches near-zero to slightly positive values, while VH remains mildly negative, indicating temporary vertical neutrality with continued lateral bias.
- **June 2025:** Both VH and VV display reduced amplitudes, with VH fluctuating between –5 mm and –14 mm and VV stabilizing near –5 mm, suggesting partial stabilization toward the end of the monitoring period.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.996258369964448, 74.7313118361176)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** VV exhibits a clear upward trend over time, transitioning from negative values (–15 mm) in January to consistently positive values (up to ~+12 mm) by May–June 2025.
- **Horizontal (VH) displacement** VH shows moderate variability with both positive and negative excursions, generally ranging between approximately –8 mm and +18 mm.

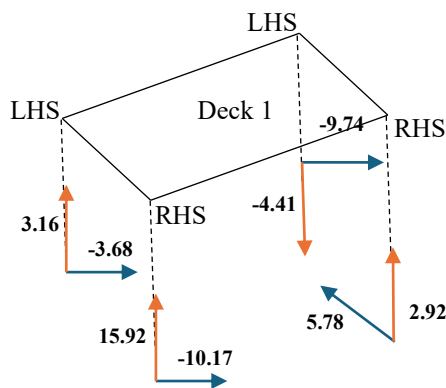
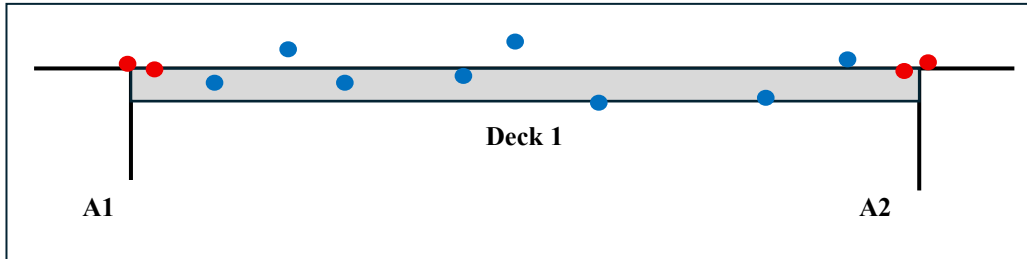
Critical Events & Anomalies:

- **January:** VV remains significantly negative while VH stays mildly positive, indicating initial settlement-dominated behaviour with limited lateral influence.
- **May 2025:** VV increases steadily to +8 to +10 mm, while VH stabilizes between +4 mm and +8 mm, indicating coordinated vertical uplift with moderate lateral stability.
- **June 2025:** Both VH and VV converge toward moderate positive values, suggesting overall stabilization.

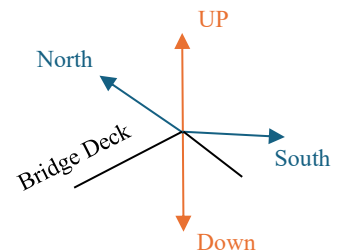
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits moderate and non-uniform vertical displacements, with a combination of uplift and settlement across LHS and RHS supports. The central lower region (RHS) records a notable uplift of approximately +15.92 mm, representing the maximum vertical movement in this configuration.

Horizontal Displacement Behaviour:

- Horizontal displacements are significant and spatially variable, dominating the overall structural response. At the deck level near RHS, a horizontal displacement of approximately -9.74 mm is observed, indicating lateral movement toward the southward direction. The central lower region shows a pronounced horizontal displacement of approximately -10.17 mm, highlighting increased lateral flexibility or reduced restraint at this support.

CONCLUSION:

- The bridge deck slab demonstrates moderate vertical movements, with localized uplift at lower regions and minor settlement at deck level. Horizontal displacements remain dominant, exhibiting opposing directions between LHS and RHS, indicative of torsional behaviour and differential boundary conditions.